

Bridge Day 13 INSTRUCTOR KEY

Loop Design, Continue, Break, and Nested State

Learning sequence: Predict - Trace - Explain - Test - Interpret

Wednesday, July 22, 2026 • 20 points • target 19 minutes

Name		UIN		Section	
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Assessment boundary and evidence source

Contract: 07a04826c975

Cumulative foundations: for/range, while, continue, break, nested state/reset

Scaffold level: complete skip/stop accumulation program

Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.18, 18.19, 18.20, 18.21, 18.22.

Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Trace nested skip, stop, reset, and accumulation. - 5 points

```
overall = 0

for row in range(2):
    row_total = 0
    for value in [0, row + 2, 5]:
        if value == 0:
            continue
        if value == 5:
            break
        row_total = row_total + value
    overall = overall + row_total
    print(row, row_total)

print(overall)
```

Question: Give a reached-pass ledger and explain exactly which loop break ends.

Model response: Row 0 resets row_total, skips 0, adds 2, and breaks at 5; it prints 0 2. Row 1 resets, skips 0, adds 3, and breaks at 5; it prints 1 3. overall becomes 5 and prints 5. break ends only the inner loop.

Item 2. Repair an accumulation/reset defect. - 3 points

```
hits = 0

for row in range(2):
    row_total = 0
    for value in [1, 2]:
        row_total = row_total + value
    hits = row_total

print(hits)
```

Question: The intent is to accumulate both row totals. Give actual, intended, and the minimal repair.

Model response: Each `row_total` is 3, but `hits = row_total` overwrites the earlier total, so actual output is 3. The intended total is 6. Repair with `hits = hits + row_total` after each inner loop.

Complete program - 12 points

- Attempt to read up to six integers using a for loop.
- Stop immediately when a negative value is read.
- Skip zero without adding it or counting it.
- For positive values, accumulate total and `accepted_count`.
- Print Total: and Accepted: on separate lines. Include a test with both skip and stop behavior.

Program workspace

```
Model program
total = 0
accepted_count = 0

for attempt in range(6):
    value = int(input())
    if value < 0:
        break
    if value == 0:
        continue
    total = total + value
    accepted_count = accepted_count + 1

print(f"Total: {total}")
print(f"Accepted: {accepted_count}")
```

Test input, expected result, and why this test matters

Reasoning and test evidence The negative test must occur before the zero and accumulation work because it ends repetition. `continue` skips the remaining body for zero but allows the next attempt. Inputs 3, 0, 4, -1 produce Total: 7 and Accepted: 2 without requesting two more values. Six positive values 1 through 6 produce Total: 21 and Accepted: 6.

Scoring guidance: 2 initialization/range, 2 stop placement, 2 skip placement, 3 accumulation/count, 1 exact output, 2 tests/explanation.